

program. For interfacing of the Hyper Terminal, refer EFY Oct. 2010 issue.

Software

The software for this project is given at the end of this article. It is written in 'Basic' language and compiled using BASCOM-AVR compiler. The source program is well commented and easy to understand.

Burn the finally obtained Intel hex code file into the AVR's flash memory using a suitable programmer. The microcontroller uses an 8MHz internally

generated clock. To activate, program fuse bytes as follows:

Fuse low byte = D4

Fuse high byte = 99

The source program is developed quickly with the help of the BASCOM-AVR library function. The 1wreset, 1wread and 1wwrite functions are used for checking the presence of 1-wire device, reading of the unique ID of iButton and writing to the 1-wire device, respectively. Sound function is used for generating the beep sound from the buzzer. The iButton is con-

nected to hardware interrupt pin INT0 so that any change on the pin can be detected and further processing for 1-wire done.

The program execution starts by initialising the input/output ports, the interrupt pin and its level of detection. Global interrupts are enabled and the interrupt service routine is ready to be executed when any interrupt is received. As soon as the normal program execution is interrupted, an interrupt is issued and the control of execution enters the interrupt service routine, where iButton is given reset command first and then the ROM command to fetch the unique ID of the iButton. This fetched ID is compared with the unique ID programmed in the firmware. If there is a mismatch between the two IDs, a fake parameter counter is incremented and the relay connected to port PD4 de-energises. If the two IDs match, the relay connected to port PD4 energises and simultaneously the buzzer connected to port PD6 sounds. If the unique ID is wrong sequentially five times, the system enters the major warning state where the buzzer sounds continuously and it can only be stopped by resetting the system.

EFY note. The source code of this project has been included in this month's EFY-CD and is also available on www.efymag.com website.

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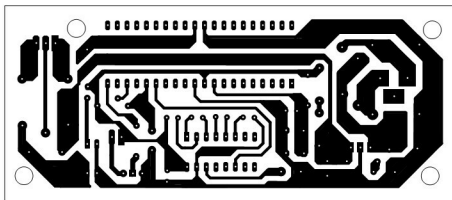


Fig. 5: An actual-size, single-side PCB for the secure digital access system using iButton

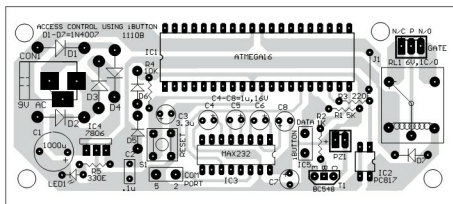


Fig. 6: Component layout for the PCB

1WIRERESET.BAS

```
Sregfile = "m16def.dat"
Sbaud = 9600
Scrystal = 8000000
$hwstack = 32
' default use 32 for the hardware stack
$swstack = 64
' default use 10 for the SW stack
$framesize = 20
' default use 10 for the frame space
Config PortD = Output
```

```
$lib "mcsbyte.lib"
Config Com1 = 9600, Synchronous = 0,
Parity = None, Stopbits = 1, Databits =
8, Clockpol = 0
Config 1wire = PortD.2
' use this pin
Config Int0 = Low Level
Dim Ar(8) As Byte, A As Word, I As
Byte, E As Byte, W As Word
Dim Codel(8) As Byte
```

```
Dim Flag As Bit
'flag for correctness
'Code(8 Byte) = {&H01, &HDD, &H83,
&H07, &H13, &H00, &H00, &H1E}
Codel(1) = &H01
Codel(2) = &HDD
Codel(3) = &H83
Codel(4) = &H07
Codel(5) = &H13
Codel(6) = &H00
```